WORKED SOLUTION

STEP II 1998, Question 10

Mechanics • Particle on Moving Wedge

STEP II 1998, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q10

The key idea

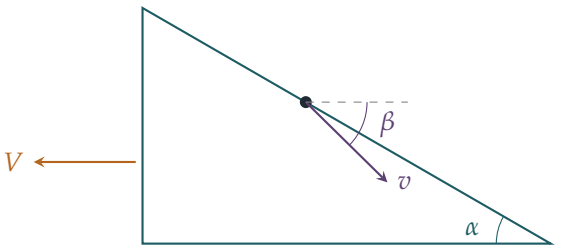
Two facts drive the whole question. Because the particle stays on the face, its velocity *relative to the wedge* must point along the face. And because nothing outside the system pushes horizontally, and nothing does work except gravity, both horizontal momentum and energy are conserved. Each of (i)–(iii) is one of these facts written as an equation. Part (ii) then shows that β never changes, which turns the final part into motion with constant acceleration.

Solution

Model. The particle is a particle; the wedge is a rigid body that slides without rotating; the face and the floor are smooth; the particle stays in contact with the face. Take the face sloping down to the right, so the particle moves down and to the right and the wedge is pushed to the left, as in the diagram.

Taking horizontal components positive to the right and vertical components positive upwards,

$$\text{particle velocity} = (v\cos\beta,\ -v\sin\beta), \qquad \text{wedge velocity} = (-V,\ 0).$$



Part (i)

The velocity of the particle relative to the wedge is

$$(v\cos\beta,\ -v\sin\beta) - (-V,\ 0) = (v\cos\beta + V,\ -v\sin\beta).$$

While the particle is on the face, this relative velocity is along the face, at angle α below the horizontal (pointing down and to the right). So

$$\frac{v\sin\beta}{v\cos\beta + V} = \tan\alpha = \frac{\sin\alpha}{\cos\alpha}.$$

Cross-multiplying,

$$\begin{aligned} & v\sin\beta\cos\alpha = v\cos\beta\sin\alpha + V\sin\alpha \\ \Longrightarrow \quad & V\sin\alpha = v(\sin\beta\cos\alpha - \cos\beta\sin\alpha) = v\sin(\beta-\alpha), \end{aligned}$$

as required. This is pure kinematics: no mechanical principle is needed, only the fact that the particle doesn't leave the face.

Part (ii)

Principle: conservation of linear momentum horizontally. Treat the particle and wedge as one system. The external forces on it are the two weights and the normal reaction from the floor, all vertical, because the floor is smooth. (The reaction between particle and face is internal.) So the total horizontal momentum is constant, and it starts at zero because everything starts from rest:

$$mv\cos\beta - MV = 0, \qquad \text{so} \qquad V = \frac{m}{M}\,v\cos\beta.$$

Substituting into (i), and expanding $\sin(\beta-\alpha)$,

$$\frac{m}{M}\,v\cos\beta\sin\alpha = v(\sin\beta\cos\alpha - \cos\beta\sin\alpha).$$

For $t > 0$ the particle is moving, so $v \neq 0$. Also $\cos\alpha \neq 0$, and $\cos\beta \neq 0$: if β were 90°, momentum would give $V = 0$, and then (i) would give $\sin(\beta-\alpha) = 0$, so $\beta = \alpha < 90^\circ$, a contradiction. Dividing by $v\cos\alpha\cos\beta$,

$$\frac{m}{M}\tan\alpha = \tan\beta - \tan\alpha \quad \Longrightarrow \quad \tan\beta = \left(1 + \frac{m}{M}\right)\tan\alpha.$$

Ask yourself Why isn't β simply equal to α?

Because the face itself is moving away to the left. Relative to the wedge the particle goes down the slope at angle α, but the wedge retreats, so relative to the ground the particle drops more steeply. Notice that the right-hand side of (ii) is a constant: β is the same at every instant, so the particle moves in a straight line. If $M \to \infty$ the wedge stays put and $\beta \to \alpha$, as it should.

Part (iii)

Principle: conservation of energy. All the contacts are smooth, so the only forces that do work on the system as a whole are the weights. (The normal reaction between particle and face does work on each body separately, but the two amounts cancel, because the relative velocity is along the face and so perpendicular to the reaction. The floor's reaction is perpendicular to the wedge's motion.) The wedge's centre of mass doesn't move vertically, so the loss of potential energy is mgy. Starting from rest,

$$mgy = \tfrac{1}{2}mv^2 + \tfrac{1}{2}MV^2.$$

Using $V = \dfrac{m}{M}v\cos\beta$ from (ii),

$$mgy = \tfrac{1}{2}mv^2 + \frac{m^2v^2\cos^2\beta}{2M}.$$

Multiplying by $2/m$,

$$2gy = v^2\Big(1 + \frac{m\cos^2\beta}{M}\Big) = \frac{v^2\,(M + m\cos^2\beta)}{M},$$

as required.

The differential equation for y

The rate at which the particle descends is the downward component of its velocity:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = v\sin\beta.$$

Put $v = \dot{y}/\sin\beta$ into (iii) (note $\sin\beta \neq 0$ since $\beta > \alpha > 0$). This gives the differential equation

$$\Big(\frac{\mathrm{d}y}{\mathrm{d}t}\Big)^2 = k\,y, \qquad \text{where} \qquad k = \frac{2gM\sin^2\beta}{M + m\cos^2\beta}.$$

The key point is that k is a *constant*, because β is constant by (ii).

To solve it, differentiate with respect to t:

$$2\dot{y}\,\ddot{y} = k\,\dot{y}.$$

For $t > 0$ the particle is descending, so $\dot{y} > 0$ and we may divide by it: $\ddot{y} = \tfrac{1}{2}k$. The particle's vertical motion has constant acceleration

$$\ddot{y} = \frac{gM\sin^2\beta}{M + m\cos^2\beta}.$$

Integrating twice with $y = 0$ and $\dot{y} = 0$ at $t = 0$ (it starts from rest),

$$y = \frac{gMt^2\sin^2\beta}{2\,(M + m\cos^2\beta)},$$

which is the required result.

(Alternatively, take the positive square root, $\dot{y} = \sqrt{ky}$, and separate: $\int y^{-1/2}\,\mathrm{d}y = \sqrt{k}\,t$ gives $2\sqrt{y} = \sqrt{k}\,t$ and the same answer. Differentiating first avoids worrying about the solution $y \equiv 0$, which also satisfies $\dot{y}^2 = ky$ with $y(0) = 0$ but isn't the motion here.)

Checks

- *Fixed wedge.* Let $M \to \infty$. Then $\beta \to \alpha$ and $y \to \frac{1}{2}gt^2 \sin^2\alpha$, which is right: on a fixed smooth slope the acceleration along the slope is $g\sin\alpha$, and its vertical component is $g\sin^2\alpha$.
- *A numerical case.* With $M = 2$, $m = 1$, $\alpha = 30^\circ$, (ii) gives $\tan\beta = \frac{3}{2}\cdot\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{2}$, so $\sin^2\beta = \frac{3}{7}$, $\cos^2\beta = \frac{4}{7}$, and the formula gives $y = \frac{1}{6}gt^2$. With $g = 9.8$ and $t = 1$ that is $y \approx 1.6333$. A direct step-by-step simulation of the wedge and particle (Newton's laws with the normal reaction found from the contact condition) gave the same value, and satisfied (i)–(iii) to within rounding in 2000 random cases.

Common slips

- Saying the particle moves at angle α to the horizontal. That is only true relative to the wedge.
- Conserving momentum in the direction of the slope, or vertically. The floor's reaction is an external vertical force, so only the horizontal component is conserved.
- Forgetting the wedge's kinetic energy $\frac{1}{2}MV^2$ in the energy equation.

Beyond STEP

A particle constrained to a moving surface is a classic test case for Lagrangian mechanics, which you meet in the first or second year at university. There you choose two coordinates (say, the wedge's position and the particle's distance down the face) and the constraint forces disappear from the equations altogether, so the constant accelerations drop out in a few lines. For Newtonian mechanics with constraints, and many problems of exactly this kind, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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